Rank distribution for a Fibonacci family

Problèmes ouverts — A. Dujella · 2026

Problem 4.10 (Dujella, 2026). Propose a heuristic to predict the rank distribution within the family of elliptic curves

$$y^2 = (F_{2k}x + 1)(F_{2k+2}x + 1)(F_{2k+4}x + 1).$$

Remark. In the range $1 \leqslant k \leqslant 100$, there are 20 values of k with rank 1, 45 values with rank 2, 21 values with rank 3, 5 values with rank between 1 and 3, 6 values with rank between 2 and 4, 2 values with rank between 3 and 5, and 1 value with rank between 2 and 6.